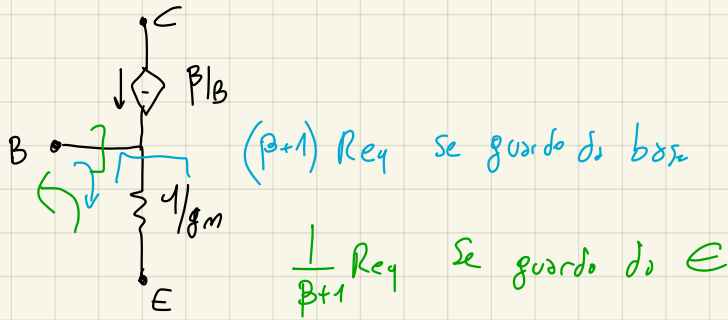
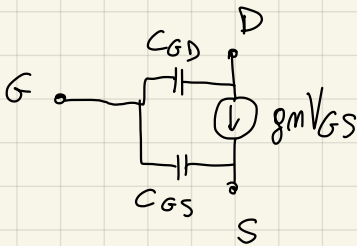


BJT

- Lo uso per generare corrente stabile



MOSFET



MATCHING TRA DISPOSITIVI

Se faccio dispositivi vicini nel wafer questi hanno circa stessi β , g_m



Con parametri P_1, P_2

$$\Delta P = P_1 - P_2$$

Sono gaussiani

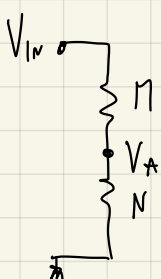
$$\sigma^2(\Delta P) = \frac{A_P^2}{A} + S_P^2 D^2$$

quadrato
distanza

Inu prop
area

Se ho **mismatching** considero questi effetti come generatori di tensione/corrente pilotati.

es particolare



$$V_A = \frac{N}{N+M} V_{IN}$$

$$\frac{\partial V_A}{V_A} = \sqrt{\frac{M}{N(N+M)}} \cdot \frac{\partial R}{R} \cdot \frac{1}{\sqrt{2}}$$

$\Rightarrow$ per fare ADC flash devo avere componenti molto precise

ANALISI IN FREQUENZA

- guadagni tensione/corrente
- impedenza / transimpedenza (e conduttanza)

$$\text{FDT generico: } \frac{Y(s)}{U(s)} = G(s) \frac{\prod (1 + s\tau_{z_i})}{\prod (1 + s\tau_{p_i})}$$

Determinazione Poli

- Numero poli = numero condensatori indipendenti

Determinazione Zeri

- $s \rightarrow \infty$, gain $\neq 0 \Rightarrow N_p = N_z$

RAGGIUNGIBILITÀ

es:
$$\begin{cases} \dot{x}_a(t) = A x_a(t) + B x_b(t) + C u(t) \\ \dot{x}_b(t) = D x_b(t) \end{cases}$$
 $\leftarrow$ Non raggiungibile dato che se non lo fa variare e nemmeno x_a (che dipende da u)

Definizione 3.1 Uno stato $\tilde{x}$ del sistema (3.92), (3.93) si dice *raggiungibile* se esistono un istante di tempo finito $\tilde{t} > 0$ e un ingresso $\tilde{u}$, definito tra 0 e $\tilde{t}$, tali che, detto $\tilde{x}_f(t)$, $0 \leq t \leq \tilde{t}$, il movimento forzato dello stato generato da $\tilde{u}$, risulti $\tilde{x}_f(\tilde{t}) = \tilde{x}$.

Un sistema i cui stati siano tutti raggiungibili si dice *completamente raggiungibile*.

es: $x_b = 0$ come movimento forzato

OSSERVABILITÀ

Definizione 3.2 Uno stato $\tilde{x} \neq 0$ del sistema (3.92), (3.93) si dice *non osservabile* se, qualunque sia $\tilde{t} > 0$ finito, detto $\tilde{y}_l(t)$, $t \geq 0$, il movimento libero dell'uscita generato da $\tilde{x}$, risulta $\tilde{y}_l(t) = 0$, $0 \leq t \leq \tilde{t}$.

Un sistema privo di stati non osservabili si dice *completamente osservabile*.

es:
$$\begin{cases} \dot{x}_a(t) = A x_a(t) \\ \dot{x}_b(t) = B x_a(t) + C x_b(t) \\ y(t) = C x_a(t) \end{cases}$$

Se $x_a(t) = 0 \Rightarrow y(t) = 0$

Indipendentemente da $x_b \Rightarrow$ parte osservabile

parte non osservabile

• Cancellazione poli/zeri

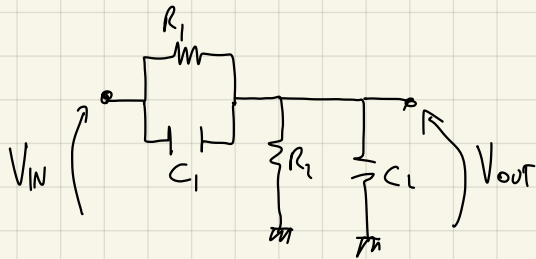
- Alcuni poli/zeri non sono osservabili/raggiungibili

Non osservabilità



Calcolo Z_{IN} , C non è osservabile (Variando tensione su condensatore non ho effetti sull'ingresso).

Non raggiungibilità



Calcolo $\frac{V_{out}}{V_{in}} \Rightarrow$ ho solo una variabile di stato

Se $\frac{C_2}{C_1} = \frac{R_1}{R_2} \Rightarrow$ Osservabile, ma non raggiungibile

$(V_{in} = V_{C1} + V_{C2})$
 $\Downarrow$
 Singolo polo

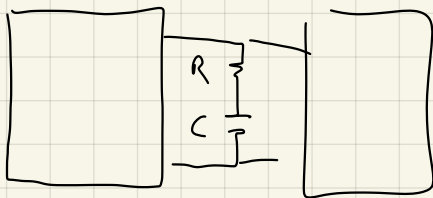
Cancellazione pol
 zero

$$V_{OUT HF} = V_{IN} \frac{\frac{1}{DC_2}}{\frac{1}{DC_1} + \frac{1}{DC_2}} = \frac{C_1}{C_1 + C_2} \neq 0$$

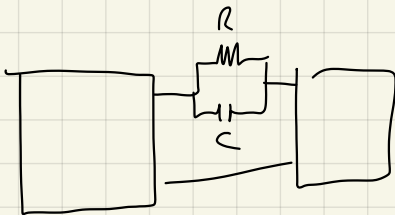
$$V_{OUT LF} = V_{IN} \frac{R_2}{R_1 + R_2}$$

- $\Rightarrow$
- $V_{HF} = V_{LF}$ Cancellazione pol zero
 - $V_{HF} > V_{LF}$ ho uno zero
 - $V_{HF} < V_{LF}$ ho un polo

CONFIGURAZIONI NOTEVOLI ZERI



$$\tau = RC$$

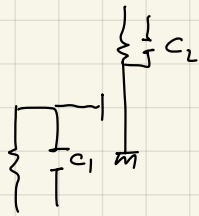


$$\tau = RC$$

CAPACITÀ INTERAGENTI

Se il variare della tensione di una $\Rightarrow$ corrente varia sull'altra

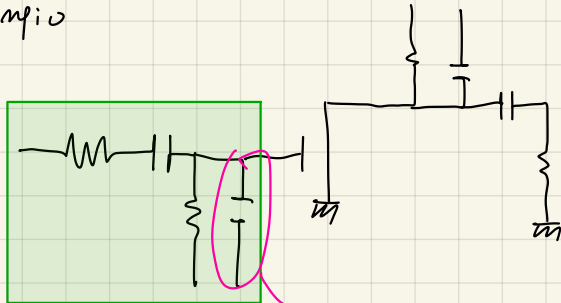
es cap non int



$$\begin{array}{l} C_1 \rightarrow C_2 \\ \cancel{C_2 \rightarrow C_1} \end{array}$$

$\Rightarrow$ Posso studiare i poli separatamente

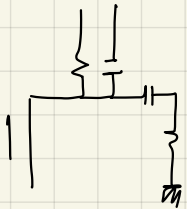
esempio



• Indipendenti

$\Downarrow$
1 zero (perché $\Delta \rightarrow \infty \Rightarrow \text{gain} = 0$ porta a massa) nell'origine

2 poli



$\Delta \rightarrow 0 \Rightarrow \text{gain} = 0 \Rightarrow$ zero in origine

2 poli

METODO DELLE COSTANTI DI TEMPO

- per risolvere capacità interagenti
- stima poli a più bassa e alta frequenza

Dato rete con capacità indipendenti e interagenti

$$\sum_{i=1}^{N_f} \tau_{p_i} = \sum R'_i C_i$$

Costanti di tempo della rete

$$\sum \omega_{p_i} = \sum_{i=1}^{N_f} \frac{1}{R''_i C_i}$$

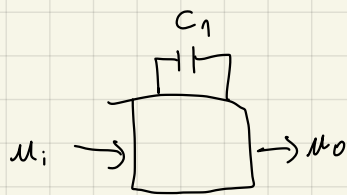
R'_i : Resistenza con altre C aperte

R''_i : Resistenza con altre C in corto

TTC method

$$H(s) = \frac{a_0 + a_1 s + \dots + a_n s^n}{b_0 + b_1 s + \dots + b_n s^n} = a_0 \frac{\left(1 - \frac{s}{z_1}\right) \left(\dots\right) \left(1 - \frac{s}{z_m}\right)}{\left(1 - \frac{s}{p_1}\right) \left(\dots\right) \left(1 - \frac{s}{p_n}\right)}$$

TTC can determine if a_i, b_i



open

$$H^0 = H|_{C_1=0}$$

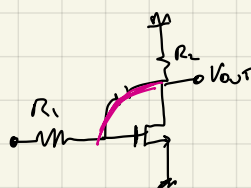
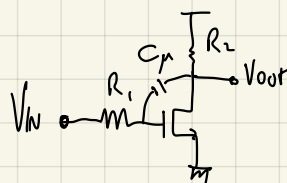
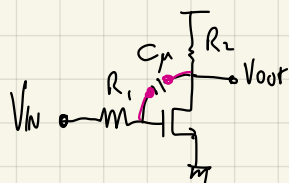
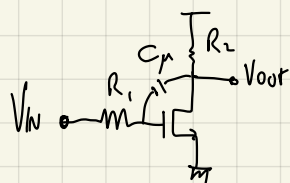
short

$$H^1 = H|_{C_1 \rightarrow \infty}$$

$$\tau = R_0 C_1$$

$$H(s) = \frac{H^0 + \tau H^1 s}{1 + \tau s}$$

es



$$H^0 = -\frac{1}{\frac{1}{g_m}} \parallel R_2 = -g_m R_2$$

$$\tau = R^0 C_\mu$$

$$= \left(\underset{\substack{\uparrow \\ s_x}}{R_1} + \underset{\substack{\uparrow \\ dx}}{R_2} + \underset{\substack{\uparrow \\ g_m \cdot s_x \cdot dx}}{g_m R_1 R_2} \right) C_\mu$$

$$H^1 = \frac{R_2 \parallel 1/g_m}{R_1 + R_2 \parallel 1/g_m}$$

ZVT

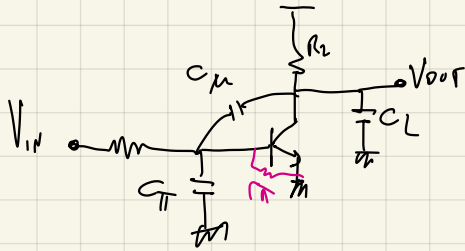
Zero Value Time constant

Determine b_1

$$b_1 = \sum_{i=1}^N \tau_i^0 \quad \left(\begin{array}{l} C \text{ open} \\ L \text{ short} \end{array} \right)$$

$$= - \sum_{i=1}^N \frac{1}{p_i}$$

- $V_{CL} = V_{CT} + V_{CM} \Rightarrow 2$ degrees of freedom
 $\Rightarrow 2$ poles



- C_T and C_L introduce zero (C_μ doesn't)

$$R_\pi^0 = R_1 \parallel r_\pi$$

$$R_\mu^0 = R_1 \parallel r_\pi + R_2 + g_m(R_1 \parallel r_\pi)R_L$$

$$R_L^0 = R_L$$

$$\tau_x^0 = C_x R_x^0$$

$$\tau_\pi^0 = 70ps$$

$$\tau_L^0 = 400ps$$

$$\tau_\mu^0 = 1200ps$$

$$\omega_h \approx \frac{1}{b_1} = \frac{1}{70 + 1200 + 400ps} = 2\pi \cdot 95MHz \quad (\text{Simulated } -3dB \approx 97MHz)$$

Attempt II: Cascode

- 4 time-constants, 3 poles (capacitive loop eliminates one initial condition)
- 1 zero, RHP (the polarity of transfer constant changes with shorting of C_μ)

$$\tau_\pi^0 = C_\pi(R_1 \parallel r_{\pi 1}) = 100fF \cdot (1k\Omega \parallel 2.5k\Omega) \approx 70ps$$

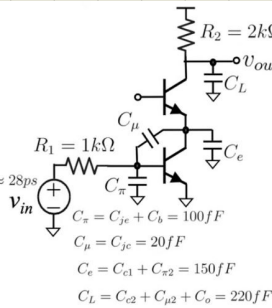
$$\tau_\mu^0 = C_\mu[R_1 \parallel r_{\pi 1} + \alpha r_{m2} + g_{m1}(R_1 \parallel r_{\pi 1})\alpha r_{m2}] \approx 20fF \cdot (710\Omega + 25\Omega + 710\Omega) \approx 28ps$$

$$\tau_e^0 = C_e \alpha r_{m2} \approx 150fF \cdot 25\Omega \approx 4ps$$

$$\tau_L^0 = C_L R_2 = 220fF \cdot 2k\Omega = 440ps$$

- Use cascode to reduce the large resistance seen by C_μ and reduce Miller multiplication
- Adds new capacitors, yet enhances the bandwidth (time constants matter, not the capacitor values per se).

$$\omega_h \approx \frac{1}{b_1} = \frac{1}{70ps + 28ps + 3.8ps + 440ps} \approx 2\pi \cdot 294MHz$$



340MHz Simulator

Attempt III: Cascode with output buffer

- 6 time-constants, 4 poles (2 capacitive loops eliminate 2 initial conditions)
- 2 zeros, one LHP (polarity doesn't change shorting $C_{\pi 3}$) and one RHP (the polarity changes with shorting of C_μ)

$$\tau_\pi^0 = C_\pi(R_1 \parallel r_{\pi 1}) = 100fF \cdot (1k\Omega \parallel 2.5k\Omega) \approx 70ps$$

$$\tau_\mu^0 = C_\mu[R_1 \parallel r_{\pi 1} + \alpha r_{m2} + g_{m1}(R_1 \parallel r_{\pi 1})\alpha r_{m2}] \approx 20fF \cdot (710\Omega + 25\Omega + 710\Omega) \approx 28ps$$

$$\tau_e^0 = C_e \alpha r_{m2} \approx 150fF \cdot 25\Omega \approx 4ps$$

$$\tau_L^0 = C_L R_2 = 220fF \cdot 2k\Omega = 440ps$$

$$\tau_{\pi 3}^0 = C_{\pi 3}(R_3 \parallel r_{\pi 3}) = 100fF \cdot (1k\Omega \parallel 2.5k\Omega) \approx 70ps$$

$$\tau_{\mu 3}^0 = C_{\mu 3}[R_3 \parallel r_{\pi 3} + \alpha r_{m4} + g_{m3}(R_3 \parallel r_{\pi 3})\alpha r_{m4}] \approx 20fF \cdot (710\Omega + 25\Omega + 710\Omega) \approx 28ps$$

$$\tau_{e 3}^0 = C_{e 3} \alpha r_{m4} \approx 150fF \cdot 25\Omega \approx 4ps$$

$$\tau_{L 3}^0 = C_{L 3} R_4 = 220fF \cdot 2k\Omega = 440ps$$

$$\tau_{L 4}^0 = C_{L 4} R_5 = 220fF \cdot 2k\Omega = 440ps$$

$$\tau_{L 5}^0 = C_{L 5} R_6 = 220fF \cdot 2k\Omega = 440ps$$

$$\omega_h \approx \frac{1}{b_1} = \frac{1}{70ps + 28ps + 4ps + 180ps + 2.5ps + 7ps} \approx 2\pi \cdot 546MHz$$

$$\omega_h \approx \frac{1}{b_1} = \frac{1}{70ps + 28ps + 4ps + 180ps + 2.5ps + 7ps} \approx 2\pi \cdot 546MHz$$

$$\omega_h \approx \frac{1}{b_1} = \frac{1}{70ps + 28ps + 4ps + 180ps + 2.5ps + 7ps} \approx 2\pi \cdot 546MHz$$

$$\omega_h \approx \frac{1}{b_1} = \frac{1}{70ps + 28ps + 4ps + 180ps + 2.5ps + 7ps} \approx 2\pi \cdot 546MHz$$

$$\omega_h \approx \frac{1}{b_1} = \frac{1}{70ps + 28ps + 4ps + 180ps + 2.5ps + 7ps} \approx 2\pi \cdot 546MHz$$

$$\omega_h \approx \frac{1}{b_1} = \frac{1}{70ps + 28ps + 4ps + 180ps + 2.5ps + 7ps} \approx 2\pi \cdot 546MHz$$

$$\omega_h \approx \frac{1}{b_1} = \frac{1}{70ps + 28ps + 4ps + 180ps + 2.5ps + 7ps} \approx 2\pi \cdot 546MHz$$

$$\omega_h \approx \frac{1}{b_1} = \frac{1}{70ps + 28ps + 4ps + 180ps + 2.5ps + 7ps} \approx 2\pi \cdot 546MHz$$

$$\omega_h \approx \frac{1}{b_1} = \frac{1}{70ps + 28ps + 4ps + 180ps + 2.5ps + 7ps} \approx 2\pi \cdot 546MHz$$

$$\omega_h \approx \frac{1}{b_1} = \frac{1}{70ps + 28ps + 4ps + 180ps + 2.5ps + 7ps} \approx 2\pi \cdot 546MHz$$

$$\omega_h \approx \frac{1}{b_1} = \frac{1}{70ps + 28ps + 4ps + 180ps + 2.5ps + 7ps} \approx 2\pi \cdot 546MHz$$

$$\omega_h \approx \frac{1}{b_1} = \frac{1}{70ps + 28ps + 4ps + 180ps + 2.5ps + 7ps} \approx 2\pi \cdot 546MHz$$

$$\omega_h \approx \frac{1}{b_1} = \frac{1}{70ps + 28ps + 4ps + 180ps + 2.5ps + 7ps} \approx 2\pi \cdot 546MHz$$

$$\omega_h \approx \frac{1}{b_1} = \frac{1}{70ps + 28ps + 4ps + 180ps + 2.5ps + 7ps} \approx 2\pi \cdot 546MHz$$

$$\omega_h \approx \frac{1}{b_1} = \frac{1}{70ps + 28ps + 4ps + 180ps + 2.5ps + 7ps} \approx 2\pi \cdot 546MHz$$

$$\omega_h \approx \frac{1}{b_1} = \frac{1}{70ps + 28ps + 4ps + 180ps + 2.5ps + 7ps} \approx 2\pi \cdot 546MHz$$

$$\omega_h \approx \frac{1}{b_1} = \frac{1}{70ps + 28ps + 4ps + 180ps + 2.5ps + 7ps} \approx 2\pi \cdot 546MHz$$

$$\omega_h \approx \frac{1}{b_1} = \frac{1}{70ps + 28ps + 4ps + 180ps + 2.5ps + 7ps} \approx 2\pi \cdot 546MHz$$

$$\omega_h \approx \frac{1}{b_1} = \frac{1}{70ps + 28ps + 4ps + 180ps + 2.5ps + 7ps} \approx 2\pi \cdot 546MHz$$

$$\omega_h \approx \frac{1}{b_1} = \frac{1}{70ps + 28ps + 4ps + 180ps + 2.5ps + 7ps} \approx 2\pi \cdot 546MHz$$

$$\omega_h \approx \frac{1}{b_1} = \frac{1}{70ps + 28ps + 4ps + 180ps + 2.5ps + 7ps} \approx 2\pi \cdot 546MHz$$

$$\omega_h \approx \frac{1}{b_1} = \frac{1}{70ps + 28ps + 4ps + 180ps + 2.5ps + 7ps} \approx 2\pi \cdot 546MHz$$

$$\omega_h \approx \frac{1}{b_1} = \frac{1}{70ps + 28ps + 4ps + 180ps + 2.5ps + 7ps} \approx 2\pi \cdot 546MHz$$

$$\omega_h \approx \frac{1}{b_1} = \frac{1}{70ps + 28ps + 4ps + 180ps + 2.5ps + 7ps} \approx 2\pi \cdot 546MHz$$

$$\omega_h \approx \frac{1}{b_1} = \frac{1}{70ps + 28ps + 4ps + 180ps + 2.5ps + 7ps} \approx 2\pi \cdot 546MHz$$

$$\omega_h \approx \frac{1}{b_1} = \frac{1}{70ps + 28ps + 4ps + 180ps + 2.5ps + 7ps} \approx 2\pi \cdot 546MHz$$

$$\omega_h \approx \frac{1}{b_1} = \frac{1}{70ps + 28ps + 4ps + 180ps + 2.5ps + 7ps} \approx 2\pi \cdot 546MHz$$

$$\omega_h \approx \frac{1}{b_1} = \frac{1}{70ps + 28ps + 4ps + 180ps + 2.5ps + 7ps} \approx 2\pi \cdot 546MHz$$

$$\omega_h \approx \frac{1}{b_1} = \frac{1}{70ps + 28ps + 4ps + 180ps + 2.5ps + 7ps} \approx 2\pi \cdot 546MHz$$

$$\omega_h \approx \frac{1}{b_1} = \frac{1}{70ps + 28ps + 4ps + 180ps + 2.5ps + 7ps} \approx 2\pi \cdot 546MHz$$

$$\omega_h \approx \frac{1}{b_1} = \frac{1}{70ps + 28ps + 4ps + 180ps + 2.5ps + 7ps} \approx 2\pi \cdot 546MHz$$

$$\omega_h \approx \frac{1}{b_1} = \frac{1}{70ps + 28ps + 4ps + 180ps + 2.5ps + 7ps} \approx 2\pi \cdot 546MHz$$

$$\omega_h \approx \frac{1}{b_1} = \frac{1}{70ps + 28ps + 4ps + 180ps + 2.5ps + 7ps} \approx 2\pi \cdot 546MHz$$

$$\omega_h \approx \frac{1}{b_1} = \frac{1}{70ps + 28ps + 4ps + 180ps + 2.5ps + 7ps} \approx 2\pi \cdot 546MHz$$

$$\omega_h \approx \frac{1}{b_1} = \frac{1}{70ps + 28ps + 4ps + 180ps + 2.5ps + 7ps} \approx 2\pi \cdot 546MHz$$

$$\omega_h \approx \frac{1}{b_1} = \frac{1}{70ps + 28ps + 4ps + 180ps + 2.5ps + 7ps} \approx 2\pi \cdot 546MHz$$

$$\omega_h \approx \frac{1}{b_1} = \frac{1}{70ps + 28ps + 4ps + 180ps + 2.5ps + 7ps} \approx 2\pi \cdot 546MHz$$

$$\omega_h \approx \frac{1}{b_1} = \frac{1}{70ps + 28ps + 4ps + 180ps + 2.5ps + 7ps} \approx 2\pi \cdot 546MHz$$

$$\omega_h \approx \frac{1}{b_1} = \frac{1}{70ps + 28ps + 4ps + 180ps + 2.5ps + 7ps} \approx 2\pi \cdot 546MHz$$

$$\omega_h \approx \frac{1}{b_1} = \frac{1}{70ps + 28ps + 4ps + 180ps + 2.5ps + 7ps} \approx 2\pi \cdot 546MHz$$

$$\omega_h \approx \frac{1}{b_1} = \frac{1}{70ps + 28ps + 4ps + 180ps + 2.5ps + 7ps} \approx 2\pi \cdot 546MHz$$

$$\omega_h \approx \frac{1}{b_1} = \frac{1}{70ps + 28ps + 4ps + 180ps + 2.5ps + 7ps} \approx 2\pi \cdot 546MHz$$

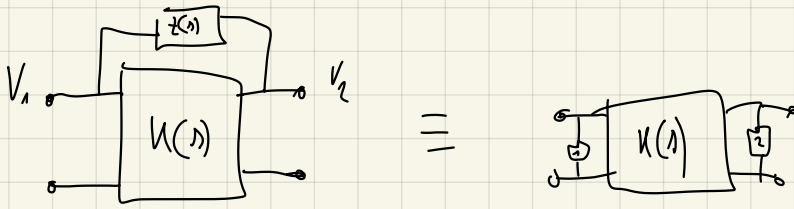
$$\omega_h \approx \frac{1}{b_1} = \frac{1}{70ps + 28ps + 4ps + 180ps + 2.5ps + 7ps} \approx 2\pi \cdot 546MHz$$

$$\omega_h \approx \frac{1}{b_1} = \frac{1}{70ps + 28ps + 4ps + 180ps + 2.5ps + 7ps} \approx 2\pi \cdot 546MHz$$



TEOREMA DI MILLER

L'impedenza $Z(s)$ vista tra $V_1(s)$ e $V_2(s)$ può essere vista con 2 impedenze



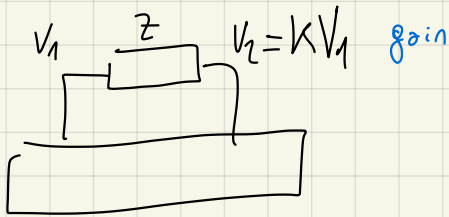
$$Z_1(s) = \frac{Z(s)}{1-K(s)}$$

$$Z_2(s) = \frac{-K(s)Z(s)}{1-K(s)}$$

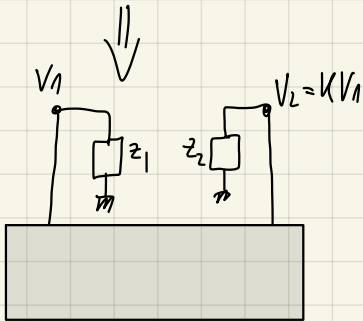
$K(s)$ è una stima, dato che dipende da V_1 e V_2

Esempio pratico

Impedenze collegate tra due nodi $\rightarrow$ 2 impedenze collegate a ground

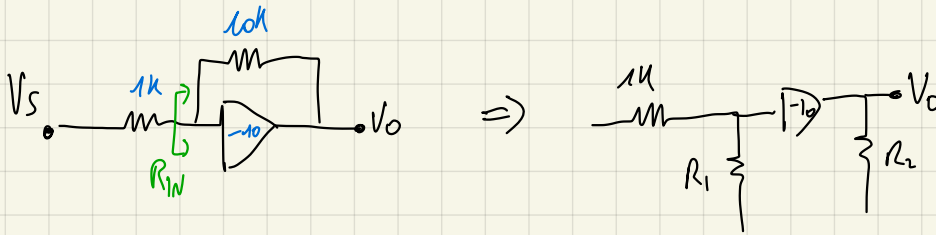


$$I = \frac{V_1 - V_2}{Z} = V_1 \frac{(1-K)}{Z}$$



$$Z_1 = \frac{Z}{1-K}$$

$$Z_2 = \frac{Z}{1 - \frac{1}{K}}$$



$$R_1 = \frac{R}{1-K}$$

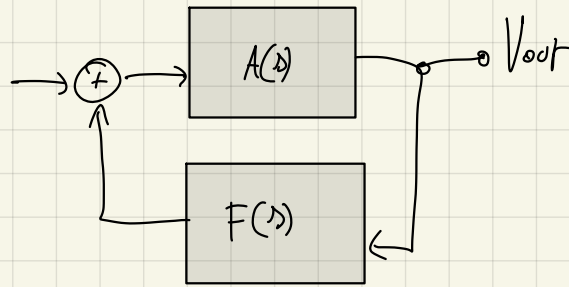
$$= \frac{10k}{11}$$

$$= 909 \Omega$$

$$R_2 = \frac{R}{1 - \frac{1}{K}} = \frac{10k}{1 - \frac{1}{-10}} = 9,1k$$

SISTEMI RETROALIMENTATI

Circuiti $\rightarrow$ Doppio bipolo



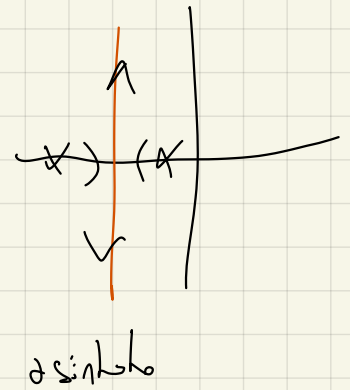
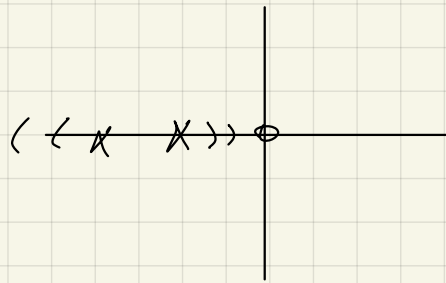
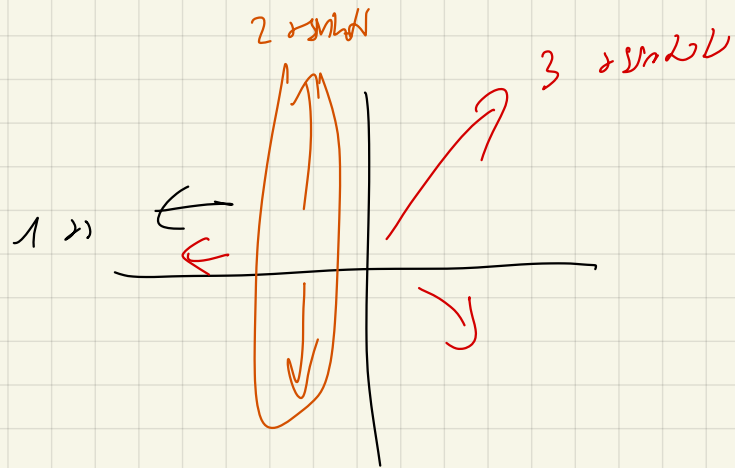
P20

DONAND E RIC

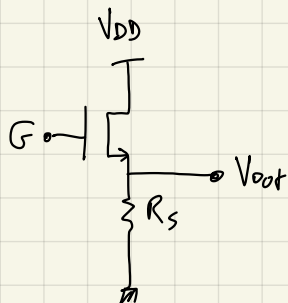
• Specchio grande ecc come si usano?

R₂ di c

1) Asintoti



COMMON DRAIN



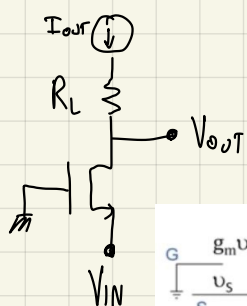
$$\frac{I_{out}}{I_{IN}} = \infty$$

$$R_{IN} = \frac{V_{IN}}{I_{IN}} = \infty$$

$$\frac{V_{out}}{V_{in}} = \frac{g_m R_S}{g_m R_S + 1} \approx 1$$

$$R_{out} = R_S // 1/g_m$$

COMMON GATE

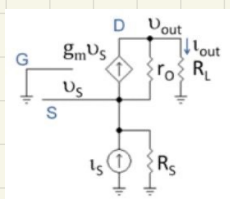


$$A_i = \frac{I_{out}}{I_S} \Big|_{R_L=0} = 1$$

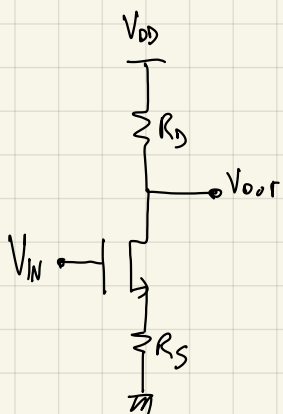
$$R_{IN} = \frac{V_S}{I_S} = \frac{R_L + r_o}{(g_m + g_{mb})r_o + 1} \approx \frac{1}{g_m}$$

$$A_V = \frac{V_{out}}{V_S} \Big|_{R_L=\infty} \approx g_m r_o$$

$$R_{out} = \frac{V_x}{I_x} = (1 + (g_m + g_{mb})r_o)R_S + r_o \approx (g_m r_o)R_S$$



COMMON SOURCE



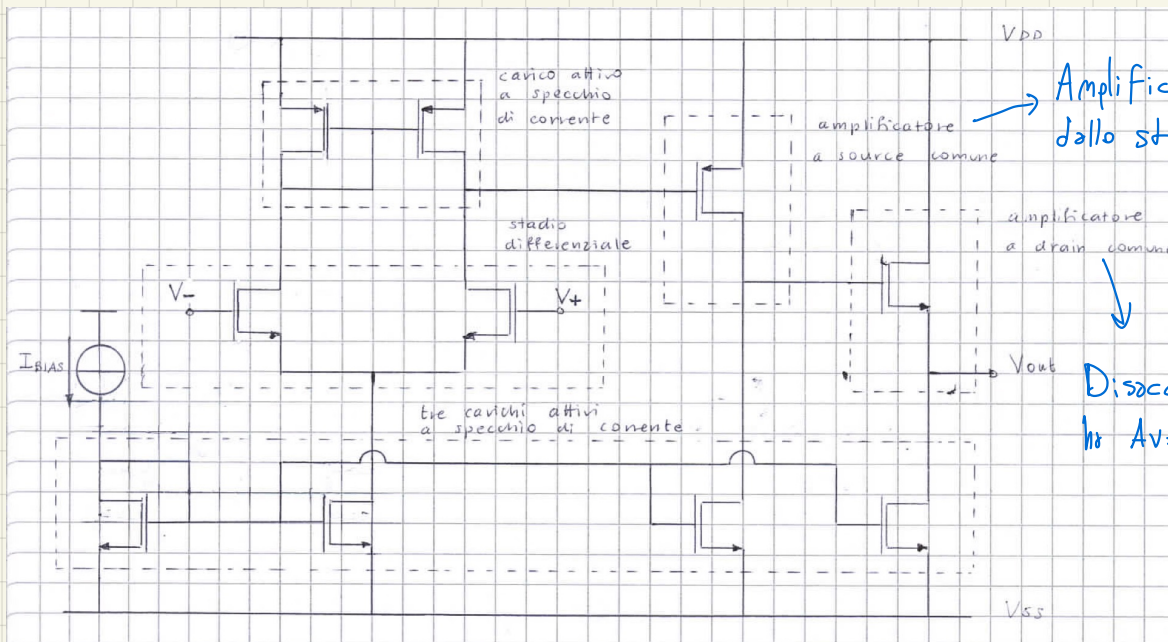
$$A_i = \frac{I_{out}}{I_{IN}} = \infty$$

$$A_V = - \frac{g_m R_D}{1 + g_m R_S}$$

$$R_{IN} = \infty$$

$$R_{out} = R_D$$

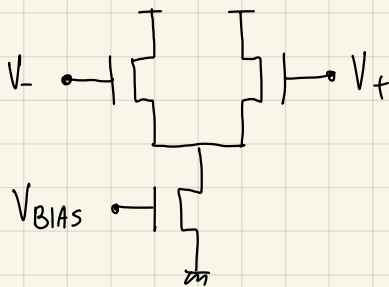
PROGETTO DI OPAMP



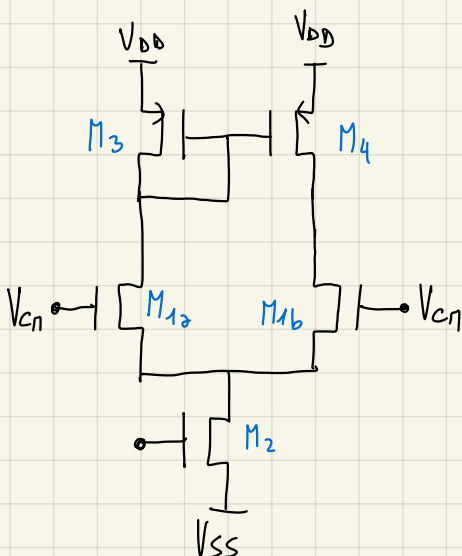
Amplifico tensione di uscita dallo stadio differenziale

Disaccoppio l'ampl e l'uscita
 $\Rightarrow Av=1$ e $R_{out} \approx 1/g_m$

STADIO DIFFERENZIALE



• ICNRR



$$V_{cn} = V_{GS1} + V_{DS2} + V_{SS}$$

$$V_{GS1} = V_{GS1a} = V_{GS1b}$$

$$\text{Saturation} \rightarrow V_{cn} \geq V_T + (V_{GS2} - V_T) + V_{SS} = V_T + V_{ov2} + V_{SS}$$

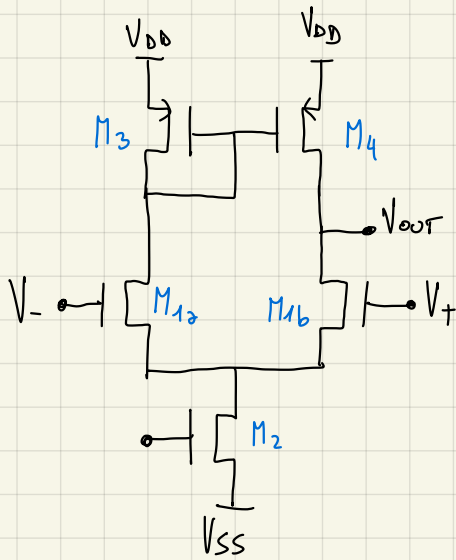
$$\Rightarrow V_{cn} = \begin{cases} V_{GS1} - V_{DS1a} - V_{SD3} + V_{DD} \\ V_{GS1} - V_{DS1b} - V_{SD4} + V_{DD} \end{cases}$$

$$V_{SD3} = V_{SG3} \geq V_T \quad V_{SD4} \geq V_{SG4} - V_T = V_{SG3} - V_T$$

$$\Rightarrow V_{cn} \leq \begin{cases} V_{GS1} + (-V_{GS1} + V_T) - V_T + V_{DD} = V_{DD} \\ V_{GS1} + (-V_{GS1} + V_T) + (-V_{GS3} + V_T) + V_{DD} = V_{DD} + V_T - V_{ov3} \end{cases}$$

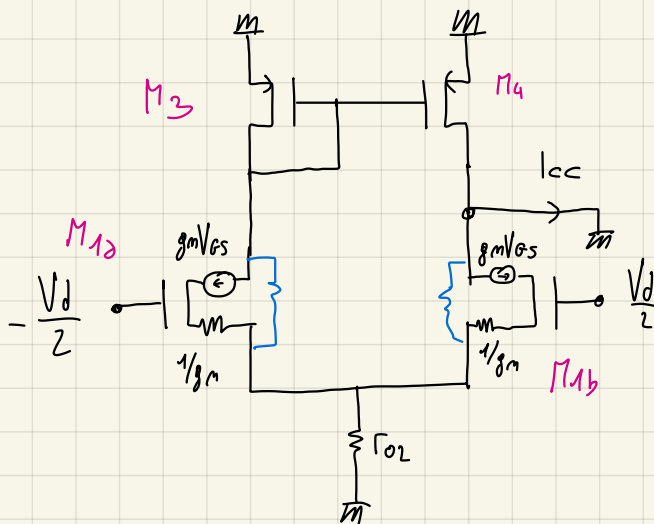
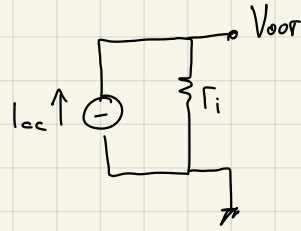
$$V_{cn} \in [V_T + V_{ov2} + V_{SS}, V_{DD}]$$

• Circuito equivalente di piccolo segnale



$$V_+ = -V_- = \frac{V_d}{2}$$

• Equivalente Norton



• Trascurate

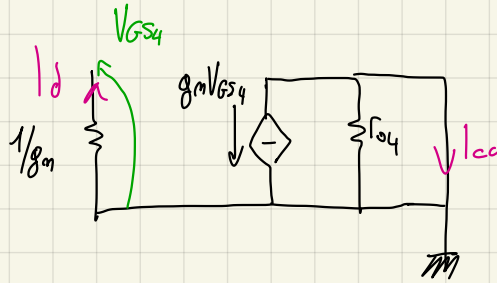
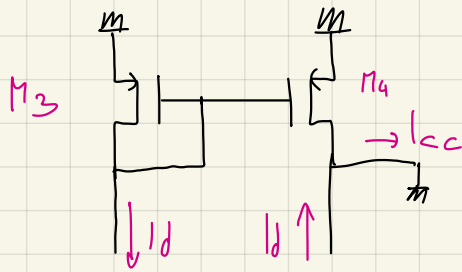
$$I_{DS1a} = \frac{-\frac{V_d}{2}}{\frac{1}{g_m} + r_{o2} \parallel \frac{1}{g_m}}$$

$$I_{DS1b} = \frac{\frac{V_d}{2}}{\frac{1}{g_m} + r_{o2} \parallel \frac{1}{g_m}}$$

$$I_{ro2} = (I_{DS1a} + I_{DS1b}) \frac{1/g_m}{1/g_m + r_{o2}} = 0 = \underbrace{I_{DS1a} + I_{DS1b}}_{\text{Current: differenziale}}$$

$$I_{DS1a} - I_{DS1b} = \frac{-V_d}{\frac{1}{g_m} + r_{o2} \parallel \frac{1}{g_m}} \approx -\frac{g_m V_d}{2}$$

Usando piccola segnale per M_3 ed M_4



• r_{o4} irrilevante • $V_{GS4} = -\frac{id}{g_m} \Rightarrow I_{CC} = id - g_m V_{GS4} = id + id = -g_m V_D$

Resistenza interna